

18.3.4 Reducing Buffer Input Resistance

A simple solution to increasing the effective bandwidth of the transmission channel is to reduce the input resistance of the input buffer. The time constant associated with the channel in Fig. 18.30 is 10 ns. When we connect the buffer to the transmission channel, the time constant drops to 3.33 ns (the three 1k resistors in parallel multiplied by the 10 pF capacitance). The drawbacks of this approach are increased power dissipation and reduced input signal amplitudes. Figure 18.31 shows how the buffer in Fig. 18.30 behaves with the input signals seen in Fig. 18.26a. The distortion (difference in the pulse widths of ones and zeroes when comparing the communication channel input, *In*, and the buffer output, *Out*) is much better.

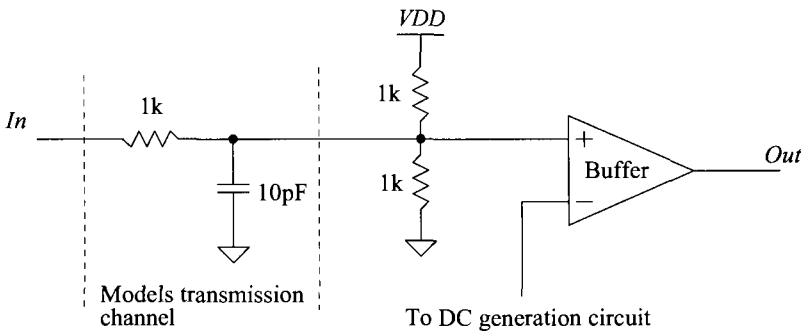


Figure 18.30 Reducing the input resistance of the input buffer.

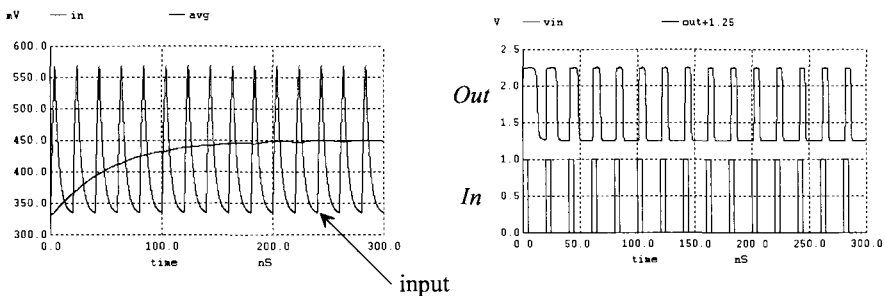


Figure 18.31 How reducing the input resistance of the buffer reduces the input signal amplitude (bad) and distortion (good).

Figure 18.32 shows an alternative low input resistance input buffer. When the input goes high, M1 turns on and M4 shuts off. This causes M2 to turn on and M5 to shut off. The output then goes high. When the input goes low, M1 turns off and M4 turns on (with M2 shutting off and M5 turning on). The result: the output goes low. Again, the power burned by M1 and M4 can be high if the transistors aren't sized properly. Further, the input swing is limited to a threshold voltage away from the supply voltages. This reduces the input voltage swing and further enhances the speed.

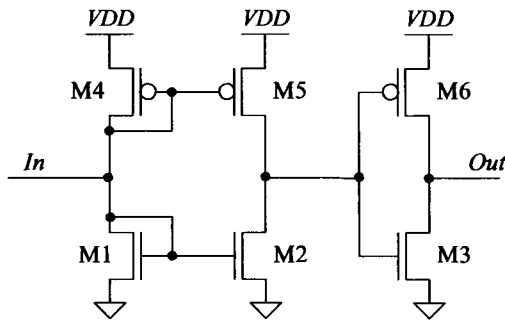


Figure 18.32 A low-input resistance input buffer.

18.4 Charge Pumps (Voltage Generators)

Often, when designing CMOS circuits, positive and negative DC voltages are needed that do not lie between ground and V_{DD} . An example of an application where a larger DC voltage source is needed was seen in Ch. 16 when we discussed Flash memory (see Fig. 16.59). A simple circuit, sometimes called a voltage pump (or more often, a *charge pump*), useful in generating a voltage greater than V_{DD} , is seen in Fig. 18.33.

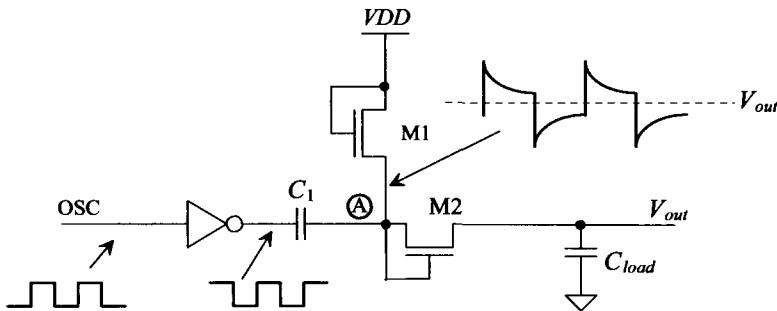


Figure 18.33 Pump used to generate a voltage greater than V_{DD} .

The operation of the voltage pump of Fig. 18.33 can be explained by first realizing that both M1 and M2 operate like a diode. M1 is simply used to pull point A to a voltage of $V_{DD} - V_{THN}$. M2 allows the charge from C_1 to charge C_{load} but not vice-versa. Let's begin the description of the circuit operation by assuming that the output of the inverter is low and point A is at a potential of $V_{DD} - V_{THN}$. When the output of the inverter goes high, the potential at point A increases to $V_{DD} + (V_{DD} - V_{THN}) = 2 \cdot V_{DD} - V_{THN}$. This turns on M2 and charges C_{load} to $2 \cdot (V_{DD} - V_{THN})$, [an extra V_{THN} because of M2's gate-source voltage drop], provided $C_1 \gg C_{load}$ and the oscillator frequency allows the capacitors to fully charge or discharge before changing states. In most practical situations, C_{load} and C_1 are comparable in size, and the output of the pump is loaded with a

DC load. The result is an output voltage with a startup time; that is, V_{out} does not immediately rise to $2 \cdot (V_{DD} - V_{THN})$ but requires several oscillator cycles to reach steady state. Also, the output has a ripple dependent on the DC load. Figure 18.34 shows the simulation results for the circuit of Fig. 18.13 using 10/1 NMOS (scale factor of 50 nm) with $C_{load} = C_1 = 1$ pF and an oscillator frequency of 10 MHz. Using this simple pump with a DC load, such as a resistor, may require employing larger MOSFETs and capacitors to avoid excessive voltage droop.

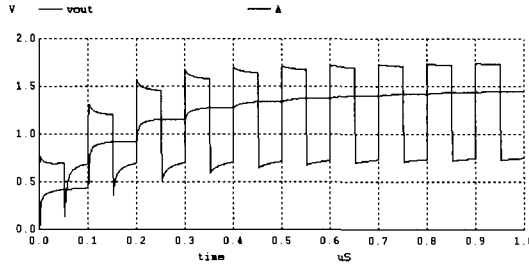


Figure 18.34 Simulating the operation of the voltage pump in Fig. 18.33.

Negative Voltages

The positive voltage pump uses n-channel MOSFETs, while the negative voltage pump, Fig. 18.35, uses p-channel MOSFETs. The reason for this comes from the requirement that the diode formed with the n+ (p+) implant used in the drain/source of the MOSFET combined with the p-substrate (n-well) does not become forward-biased. Forward biasing this parasitic diode is an *important concern* and is often the reason why some more exotic pump topologies can't be implemented. For example, in Fig. 18.35, we connect the well of the PMOS devices to ground instead of to their respective sources. Consider what would have happened had we connected the well to the source of the MOSFET. When the output voltage goes negative, the (n-type) well goes negative too. If the substrate (p-type) is at ground, this forward biases the n-well to substrate diode acting to clamp the output of the pump at a negative diode drop. While we could have left the bodies of the PMOS (the n-wells) tied to V_{DD} , here we chose to connect them to ground so that the body effect wasn't so severe (a lower threshold voltage).

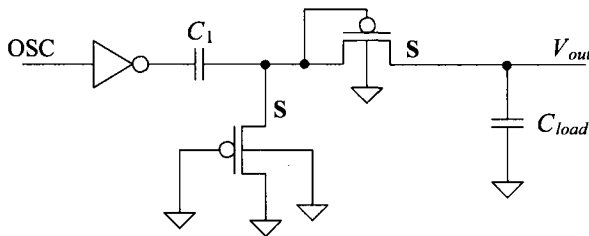


Figure 18.35 Negative voltage pump.

Using MOSFETs for the Capacitors

The capacitors used in Figs. 18.33 and 18.35 can be replaced with MOSFETs. An example of using n-channel MOSFETs in place of the capacitors in Fig. 18.33 is shown in Fig. 18.36. The main requirement on a MOSFET used as a capacitor is that its V_{GS} remain greater than V_{THN} at all possible operating conditions. In other words, the MOSFET must remain in the strong inversion region so that its capacitance is a constant $C'_{ox} \cdot W \cdot L$. The capacitor, C_{load} , in Fig. 18.36 remains in strong inversion because $V_{out} \gg V_{THN}$. The capacitor C_1 remains in strong inversion because, when the inverter output is low, the voltage on the gate of C_1 is $VDD - V_{THN}$ ($= V_{GS}$). When the output of the inverter is high (VDD), the voltage on the gate of C_1 is $2 \cdot VDD - V_{THN}$. In both cases (the output of the inverter high and low), $V_{GS} = VDD - V_{THN}$, and the MOSFET is in the strong inversion region.

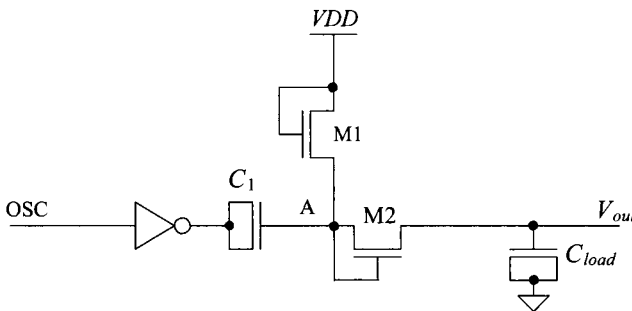


Figure 18.36 Using MOSFETs as capacitors.

18.4.1 Increasing the Output Voltage

Figure 18.37 shows a voltage pump with a higher output voltage. The increase in the output voltage comes from eliminating the threshold voltage drop at the gate and drain of M7. This allows the output to swing up to $2 \cdot VDD - V_{THN}$.

To understand the operation of the circuit, let's assume that the voltage at point A is low and the voltage at point C is $VDD - V_{THN}$. When the output of INV1 goes high, point A is VDD and the voltage at point C swings up to $2 \cdot VDD - V_{THN}$. This causes M4, M5, and M6 to turn on and pull points D and E to VDD . Now when point B goes high (point A goes back to zero), points D and E swing up to $2 \cdot VDD$ and the output goes to $2 \cdot VDD - V_{THN}$. Note that MOSFETs M2 and M3 are not needed; unless the pump drives a DC load, they never turn on. Also, separating points D and E is unnecessary unless the pump supplies a DC current.

18.4.2 Generating Higher Voltages: The Dickson Charge Pump

The voltage pumps of the last section are limited to voltages less than $2 \cdot VDD$ and greater than $-2 \cdot VDD$. Figure 18.38 shows a scheme for generating arbitrarily high voltages (limited by the breakdown voltage of the capacitors or the oxide breakdown of the MOSFETs). Again, the MOSFETs are used as diodes in this configuration.

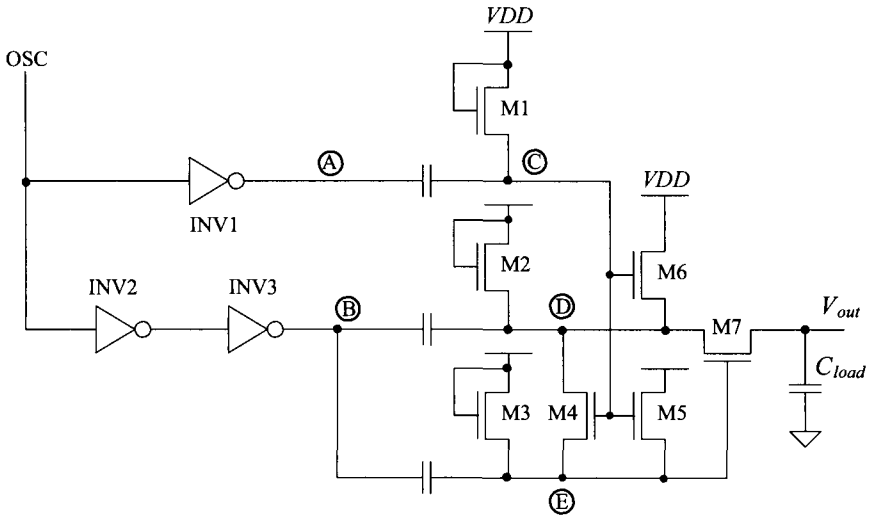


Figure 18.37 Increased output voltage pump.

In the following description of circuit operation, we assume steady-state operation with no DC load present. When CLK is low, point A is pulled, by M1, to $V_{DD} - V_{THN}$. When CLK goes high, point A swings up to $2 \cdot V_{DD} - V_{THN}$. Diode M2 turns on, and point B charges to $2 \cdot V_{DD} - 2 \cdot V_{THN}$. When CLK goes low, $\overline{\text{CLK}}$ goes high and point B swings up to $3 \cdot V_{DD} - 2 \cdot V_{THN}$. When $\overline{\text{CLK}}$ goes low, point B swings back down to $2 \cdot V_{DD} - 2 \cdot V_{THN}$. This operation proceeds through the circuit to the last stage of the multiplier. The output of the multiplier swings from $(N+1) \cdot V_{DD} - N \cdot V_{THN}$ down to $N \cdot V_{DD} - N \cdot V_{THN}$. C_N can be made larger than the other capacitors to reduce this ripple.

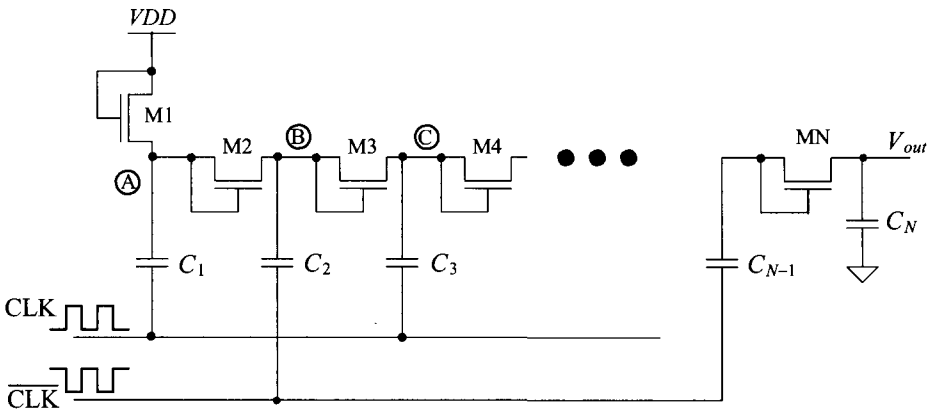


Figure 18.38 N-stage voltage multiplier (the Dickson charge pump).

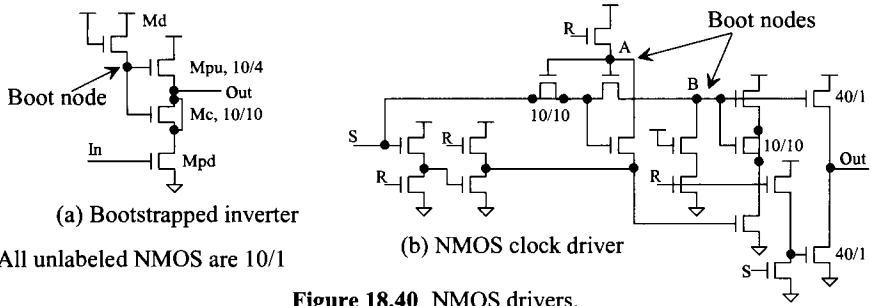


Figure 18.40 NMOS drivers.

18.4.3 Example

A common application of a voltage generator in digital circuits is generating a negative substrate bias; that is, instead of tying the substrate to ground, the substrate is held at some negative voltage. Typically, this negative voltage is between -0.5 and -1 V. “Pumping” the substrate negative is found in some DRAMs. A negative substrate bias has several benefits. It (1) stabilizes n-channel threshold voltages, (2) increases latch-up immunity after power up, (3) prevents forward biasing n+ to p-substrate pn junction, (4) allows chip inputs to go negative without forward biasing a pn junction, (5) prevents substrate from going locally above ground, (6) reduces depletion capacitances associated with the n+ to p-substrate junction, and (7) reduces subthreshold leakage current.

A simple substrate pump is shown in Fig. 18.41. In this circuit we can use n-channel MOSFETs to generate a negative potential, unlike Fig. 18.35, since the negative voltage is connected to the substrate. In this situation, the drain/source implants of the n-channel MOSFETs cannot become forward biased. Note the absence of the load capacitance. It turns out that the capacitance of the substrate to everything else in the circuit presents a very large capacitance to the pump. In other words, the substrate itself is a very large capacitor.

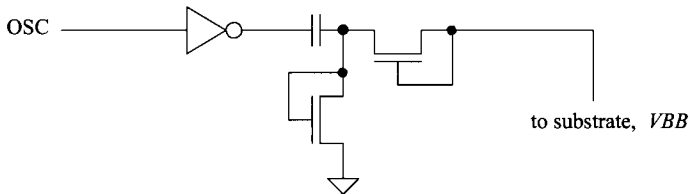


Figure 18.41 Simple substrate pump.

An example oscillator circuit that is used to drive the substrate pump is shown in Fig. 18.42. This is a standard ring oscillator with a NAND gate to enable/disable the oscillator. Capacitors are added at a few points in the middle of the oscillator to increase the delay and lower the frequency of the oscillator so that the pump’s capacitors fully charge.

The final component in a substrate pump is the regulator, a circuit that senses the voltage on the substrate and enables or disables the substrate pump (the oscillator). Using

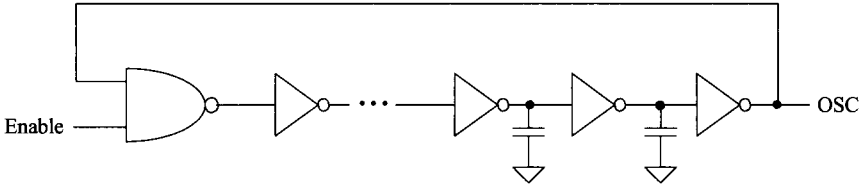


Figure 18.42 Ring oscillator with enable.

a comparator with hysteresis, a precision voltage reference, and a level shifting circuit, the enable signal can be generated with the circuit of Fig. 18.43. The hysteresis of the comparator determines the amount of ripple on the substrate voltage. Forcing a constant current through the two MOSFETs, M1 and M2, compels their source-gate voltages to remain constant (note how the body effect is eliminated by using p-channel MOSFETs), causing the MOSFETs to behave as if they were batteries. The battery action of M1 and M2 shifts the substrate voltage up so that it lies in the input range of the comparator. The number of MOSFETs (in this case two) and the magnitude of the current I determine the substrate voltage.

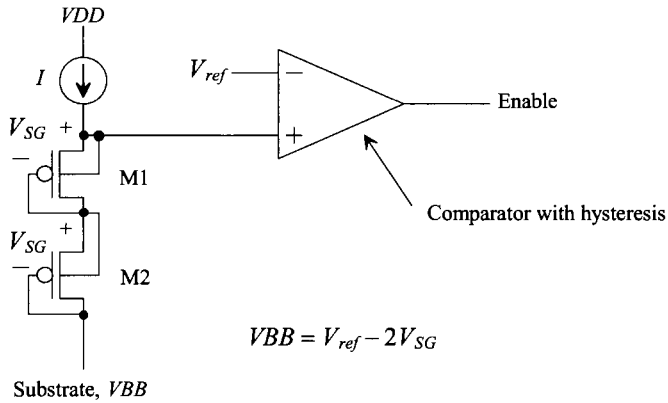


Figure 18.43 Regulator circuit used in substrate pump.

A simpler and less accurate implementation of the regulator is shown in Fig. 18.44. The comparator and voltage reference are implemented with an inverter and M1–M3. MOSFET M3 causes the inverter to have hysteresis, that is, behave like a Schmitt trigger. MOSFETs M1 and M2 form an inverter with a switching point voltage of approximately V_{THN} . The current source of Fig. 18.43 is implemented with the long L MOSFET M4 in Fig. 18.44. The level shifting is performed with the n-channel MOSFETs M5 and M6. When point A gets above a potential of V_{THN} , the Enable output goes high, causing the substrate pump to turn on and drive the substrate voltage negative. When point A gets pulled, via the substrate voltage through M5 and M6, below V_{THN} , the Enable output goes low and the pump shuts off. The substrate voltage generated with this circuit is approximately $-V_{THN}$ with body effect.

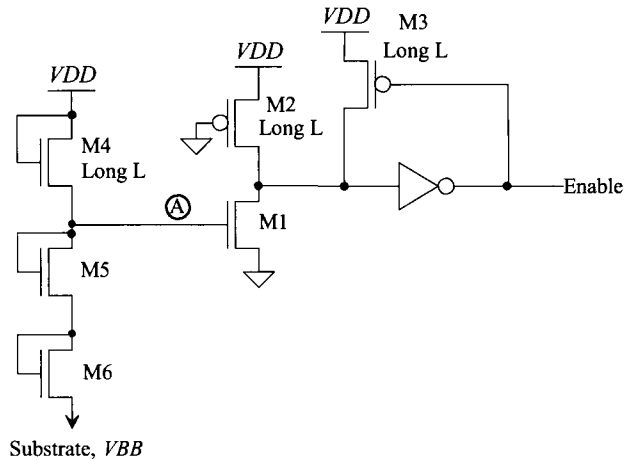


Figure 18.44 Simpler substrate regulator.

ADDITIONAL READING

- [1] R. J. Baker, "Input and output buffers having symmetrical operating characteristics and immunity from voltage variations," US Patent 7,102,932, September 5, 2006.
- [2] F. Pan and T. Samaddar, *Charge Pump Circuit Design*, McGraw-Hill, 2006. ISBN 978-0071470452.
- [3] H. Lin, N.-H. Chen, and J. Lu, "Design of Modified Four-Phase CMOS Charge Pumps for Low-Voltage Flash Memories," *Journal of Circuits, Systems, and Computers*, Vol. 11, No. 4, pp. 393-403, 2002. Excellent paper.
- [4] N. Otsuka and M. A. Horowitz, "Circuit techniques for 1.5-V power supply flash memory," *IEEE Journal of Solid-State Circuits*, Vol. 32, pp. 1217-1230, August 1997.
- [5] M. Bazes, "Two Novel Full Complementary Self-Biased CMOS Differential Amplifiers," *IEEE Journal of Solid-State Circuits*, Vol. 26, No. 2, pp. 165-168, February 1991.
- [6] J. F. Dickson, "On-Chip High-Voltage Generation in MNOS (sic) Integrated Circuits Using an Improved Voltage Multiplier Technique," *IEEE Journal of Solid State Circuits*, Vol. SC-11, No. 3, June 1976.
- [7] J. Millman and H. Taub, *Pulse, Digital, and Switching Waveforms*, McGraw-Hill Publishers, 1965. ISBN 07-042386-5. Information on multivibrators and oscillators.
- [8] O. H. Schmitt, "A Thermionic Trigger," *Journal of Scientific Instruments*, Vol. XV, pp. 24-26, No. 1, Jan. 1938. Information on the "Schmitt trigger."

PROBLEMS

For the following problems use the short-channel CMOS process with a scale factor of 50 nm and a V_{DD} of 1 V.

- 18.1** Design a Schmitt trigger with $V_{SPL} = 0.35$ and $V_{SPH} = 0.55$ V. Use SPICE to verify your design.
- 18.2** Estimate t_{PHL} and t_{PLH} for the Schmitt trigger of Ex. 18.1, driving a 100 fF load capacitance. Compare your hand calculations to simulation results.
- 18.3** Design and simulate the operation of a Schmitt trigger-based oscillator with an output frequency of 10 MHz. Simulate the design using SPICE.
- 18.4** Estimate the total input capacitance on the control voltage input for the VCO shown in Fig. 18.8.
- 18.5** Design an astable multivibrator with an output oscillation frequency of 20 MHz.
- 18.6** Design and simulate the operation of a one-shot that has an output pulse width of 100 ns. Comment on the maximum rate of retrigger and how ESD diodes connected to bonding pads will affect the circuit operation if the resistor and capacitor are bonded out.
- 18.7** Design and simulate a 100 MHz oscillator using the astable multivibrator in Fig. 18.11. Comment how process shifts in the resistor and capacitor affects the oscillation frequency. If the resistor and capacitor are bonded out, will the ESD diodes affect the circuit's operation?
- 18.8** Using the buffer seen in Fig. 18.23 to drive a load capacitance of 100 fF, plot the propagation delays against the V_{inp} input signal amplitude when V_{inm} is 250, 500, and 700 mV. (V_{inp} is centered around V_{inm} .)
- 18.9** The DC restore circuit seen in Fig. 18.28 has limitations, such as it stops working if a long string of 1s or 0s is applied to the input buffer or the input data moves too quickly and the long RC time constants keep the DC restore output from following the center of the data. Comment, with the help of SPICE simulation results, on these limitations. Show how changing the values of the resistors and capacitors can compensate for these limitations. (For example, using longer time constants allows for longer strings of ones or zeroes before the DC restore signal is invalid.)
- 18.10** Using the model for the transmission channel seen in Figs. 18.26 and 18.30, show that the circuit in Fig. 18.32 will work as an input buffer. Using SPICE show the limitations of the buffer (signal amplitude and speed). Also, comment on the power pulled by the buffer.
- 18.11** Design a nominally 2 V voltage generator that can supply at most 1 μ A of DC current (2 MEG resistor). Simulate the operation of your design with SPICE.
- 18.12** Design and simulate the operation of a nominally -1 V substrate pump. Comment on the design trade-offs.

Digital Phase-Locked Loops

The digital phase-locked loop, DPLL, is a circuit that is used frequently in modern integrated circuit design. Consider the waveform and block diagram of a communication system shown in Fig. 19.1. Digital data¹ is loaded into the shift register at the transmitting end. The data is shifted out sequentially to the transmitter output driver. At the receiving end, where the data may be analog (and, thus, without well-defined amplitudes) after passing through the communication channel, the receiver amplifies and changes the data back into digital logic levels. The next logical step in this sequence is to shift the data back into a shift register at the receiver and process the received data. However, the absence of a clock signal makes this difficult. The DPLL performs the function of generating a clock signal, which is locked or synchronized with the incoming signal. The generated clock signal of the receiver clocks the shift register and thus recovers the data. This application of a DPLL is often termed a *clock-recovery circuit* or *bit synchronization circuit*.

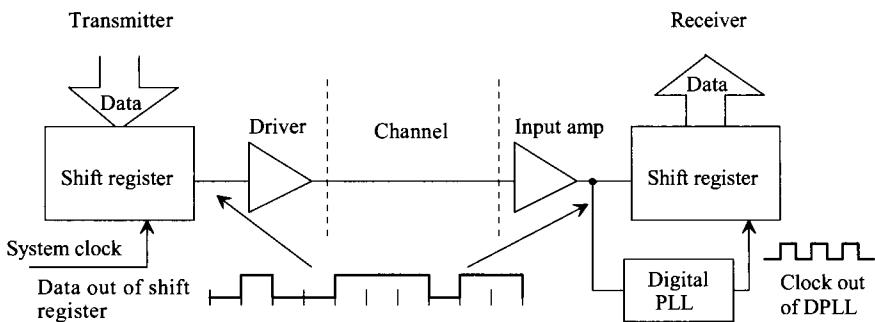


Figure 19.1 Block diagram of a communication system using a DPLL for the generation of a clock signal.

¹ While “data” is plural and “datum” is singular, we will not use, in this chapter, the grammatically correct “data are” in favor of the colloquial “data is”.